

**ECON 101—Introduction to Microeconomics
Spring 2023**

Final

April 30, 2023

Name:

NU ID number:

Instructions:

- Answer all questions. Multiple choice answers go in the space provided below. All questions have the same weight. You will be provided with scratch paper.
- The exam has a duration of 2 hours.
- You are not allowed to go out during the duration of the exam. This means no bathroom breaks. Thus, be prepared.
- You are only allowed to have in your desk area: ID, writing instruments, ruler, and a simple calculator (no graphing calculators), and a simple watch if needed (no smartwatches).
- You must leave all other personal items, including smart devices, to the side of the room. It may be a better idea to not bring them at all.
- If you finish early the exam, you will only be allowed to go out until 10:45 PM. Otherwise, you will have to remain in the room quietly until the exam is finished.
- You are not allowed to remove the staple from your exam booklet.

Multiple choice. Please write your answers here!

1	2	3	4	5	6	7	8	9	10

11	12	13	14	15

Bonus question: What is the name of Prof. Alejandro's cat? _____

MCQ (Problems 1 and 2)	Problem 3.1	Problem 3.2	Bonus	Total

Problem 1

Everybody knows that Prof. Alejandro is crazy about coffee, I mean he even uses his brew method as an example in his ECON101 class. One day, while staring at the ceiling during an unproductive day, he had an epiphany: “Why not open my own coffee stand in NU? I’ve got the beans, the brew, and the brains for it!” After a bit more thinking, he decided the most important thing, which would be to name his stand “Panqueque’s Coffee & Sweets.” However, being an economist, he knows that he needs to carefully use his knowledge of Economics to analyze the feasibility of his coffee stand before diving in!

Prof. Alejandro decides that what is missing from NU’s current coffee options is a *pour-over style brew*. You will be spared of reading the whole details¹ of the preparation method, but at its core, this style of brew uses three inputs: coffee in grams c , water in grams w , and labor in minutes ℓ . The recipe uses the inputs in *fixed proportions* as follows: every cup of coffee q is made with 15 grams of coffee, 250 grams of water, which is brewed over 5 minutes exactly by pouring a constant stream of water.

1. According to the above recipe, what would be the appropriate production function to model the pour-over style brew recipe?

- (a) $q = \left(\frac{c}{15} + \frac{w}{250} + \frac{\ell}{5} \right)^{1/3}$.
- (b) $q = c^{1/15} + w^{1/250} + \ell^{1/5}$
- (c) $q = c^{1/15} w^{1/250} \ell^{1/5}$
- (d) $q = \min \left\{ \frac{c}{15}, \frac{w}{250}, \frac{\ell}{5} \right\}$

2. For the production function modelling the recipe, what kind of returns to scale it exhibits?

- (a) Decreasing returns to scale
- (b) Increasing returns to scale
- (c) Constant returns to scale
- (d) None of the above

3. Suppose that Prof. Alejandro intends to work on his coffee stand as a side job. Thus, he will only be able to devote $\bar{\ell} = 20$ hours per week to the coffee stand, so that labor will be a *fixed input*. The period of analysis will be *1 month, or 4 weeks*. If this is the case, and assuming he has the required amounts of coffee and water, what is the maximum amount of coffee cups per month that he can brew?

- (a) 960
- (b) 1000
- (c) 550
- (d) 240

4. The next step is to figure out the cost structure of the coffee stand. The price of a bag of 1 kilogram of *single origin coffee* costs 12,000 Tenge. Bottles of water of 5 liters cost 660 Tenge. The cost of opportunity of labor per hour (its price) is 5200 Tenge. Other fixed costs (electric gooseneck kettles, weighting scales, ceramic coffee drippers, an electric coffee grinder, and the furniture for the stand) amount to 1,500,000 Tenge. What is the Short-Run Total Cost function for the stand?

- (a) $STC(q) = \frac{1939}{3}q + 1500000$
- (b) $STC(q) = 213q + 1,916,000$ for all levels of q .
- (c) $STC(q) = 213q + 1,916,000$ for $0 \leq q \leq 960$, becomes vertical at $q = 960$.
- (d) I don’t know, but everything seems very expensive!

¹Feel free to ask me later if interested, but prepare yourselves to hear a long explanation of coffee chemistry.

Prof. Alejandro realizes that if he intends to open his stand in the NU atrium, he would be entering a *very competitive market*. To simplify the analysis, he considers that he would be operating under *perfect competition*.

5. Given the cost structure derived above, what is the shutdown price, p^S , for the coffee stand?
 - (a) $p^S = 960$
 - (b) $p^S = 213$
 - (c) $p^S = \frac{1939}{3}$
 - (d) $p^S = 240$

6. Given the cost structure derived above, what is the break-even price, p^B , for the coffee stand?
 - (a) $p^B = 960$
 - (b) $p^B = 213$
 - (c) $p^B = 1500$
 - (d) $p^B = 2208.83$

7. If the price of a cup of coffee is p , what is the coffee stand's supply curve?
 - (a) Horizontal, with a value of 213.
 - (b) Vertical at $q = 240$.
 - (c) Vertical, at $q = 960$.
 - (d) Horizontal, with a value of 213, from $0 \leq q \leq 960$, becomes vertical at $q = 960$.

8. The average price of a cup of coffee at NU is 800 Tenge. At this price, *Panqueque's Coffee & Sweets* would choose to produce _____ cups of coffee, and would be making _____ in the amount of _____.
 - (a) 960, losses, $-943, 520$
 - (b) 0, neither, 0.
 - (c) 960, profits, $943, 520$
 - (d) I think it maybe a better idea that Prof. Alejandro prepares other types of beverages, like the ones that appeared in the last exam.

9. In the long-run, what would be your prediction?
 - (a) All the other coffee shops should reduce their prices so that *Panqueque's Coffee & Treats* can continue operating in the long-run.
 - (b) *Panqueque's Coffee & Treats* should continue operating in the long-run because it is able to cover its variable costs.
 - (c) Fixed costs are too high, so unless Prof. Alejandro manages to reduce his costs or work more hours in the coffee stand, *Panqueque's Coffee & Treats* will have to exit the market.
 - (d) Prof. Alejandro should focus all his time in his job as a Professor and do not daydream about opening coffee shops.

Problem 2

Consider the market for electric scooters in Astana. Suppose that this market is in a long-run equilibrium. In this equilibrium, each firm's short-run and long-run cost functions are given by:

$$\begin{aligned}SRTC(q) &= q^3 - 3q^2 + 3q + 4, & SRMC(q) &= 3q^2 - 6q + 3 \\LRTC(q) &= 3q\end{aligned}$$

The market demand for electric scooters is given by $D(P) = 27 - P$.

10. What is the equilibrium price in the initial long-run equilibrium?
 - (a) 5
 - (b) 3
 - (c) 0
 - (d) 4
11. How many electric scooters does each firm produce in the long-run equilibrium? *Hint: If the market is in a long-run equilibrium, it is also in a short-run equilibrium.*
 - (a) 2
 - (b) 3
 - (c) 1
 - (d) 0
12. What is the equilibrium market quantity in the initial long-run equilibrium and how many firms are in the market?
 - (a) $Q^* = 23$ and there are $N = 11$ firms.
 - (b) $Q^* = 24$ and there are $N = 12$ firms.
 - (c) $Q^* = 22$ and there are $N = 9$ firms.
 - (d) $Q^* = 24$ and there are $N = 10$ firms.
13. The firms' short-run supply curve and the market supply curve are:
 - (a) $q(p) = 1 + \sqrt{p/3}$, $Q(p) = 12 + 12\sqrt{p/3}$
 - (b) $q(p) = 1 + \sqrt{p/3}$ for $p \geq 3$, 0 otherwise; $Q(p) = 12 + 12\sqrt{p/3}$ for $p \geq 3$, 0 otherwise.
 - (c) $q(p) = 1 + \sqrt{p/3}$ for $p \geq 3/4$, 0 otherwise; $Q(p) = 12 + 12\sqrt{p/3}$ for $p \geq 3/4$, 0 otherwise.
 - (d) $p = 3q^2 - 6q + 3$, $p = 36Q^2 - 72Q + 36$
14. Suppose that electric scooters suddenly experience a rise in popularity, and thus, the market demand shifts to $D'(p) = 57 - 3p$. What are the equilibrium price and quantity in the short run?
 - (a) $Q^* = 54.75$, $p^* = 3/4$.
 - (b) $Q^* = 54$, $p^* = 1$.
 - (c) $Q^* = 48$, $p^* = 3$.
 - (d) $Q^* = 32$, $p^* = 25/3$.
15. If the market stays at $D'(p)$ from now on, how will the market adjust in the long run? How many firms will there be in the long run?
 - (a) $(Q^*, p^*) = (24, 3)$, $N = 12$
 - (b) $(Q^*, p^*) = (48, 3)$, $N = 24$.
 - (c) $(Q^*, p^*) = (54, 1)$, $N = 5$
 - (d) $(Q^*, p^*) = (32, 25/3)$, $N = 12$

Problem 3: True/False questions

For the next two statements, write whether it is true or false, and justify your answer. Your argument should be short, answer it in the space provided.

1. Consider the market for cars. All firms have the same production technology, and we know that the long-run average cost curve has a minimum at a cost of 4,600,000 Tenge per car. Therefore, we can never observe a market equilibrium in which the equilibrium price is below 4,600,000 Tenge per car.
2. Consider a firm that produces computers. Suppose that the market price of a computer is $\bar{p}$, both in the short-run and the long-run. If in the short-run this firm decides to shut-down, then it must be that in the long-run it decides to exit the market.

Write here anything you want me to consider about Problem 1 for partial credit.

Write here anything you want me to consider about Problem 2 for partial credit.